CONTENTS

INTRODUCTION

The objective of the automotive headlight is to enhance safety at night by improving the driver's visibility in low light and poor weather conditions. Like other important components of a vehicle, head-lights have the chief responsibility to brighten up the road to provide a clear vision a head in the direction of vehicle movement. Automotive Lighting history dates back to the 1880s. In the 1880s, the gas and oil lamps were used as a lighting source in the animal-drawn coaches. While, in 1908 the first electric bulb was lighted with Dynamo in Automotive, and the Halogen lamps were first time introduced in Europe in 1960. Later on, Xenon (HID) lamps, LEDs, and Laser source continued this research. Research and development to innovate and improve driver visibility is ongoing since one and a half-century.

Adaptive front lighting systems (AFS) attempt to dynamically adjust the headlights of the vehicle so that the driver has optimum nighttime vision without compromising the safety of other road users. The AFS uses stepper motors to control the headlight angle when the vehicle steers or the road is not even. Besides, the adaptive system tries to avoid a direct glare to oncoming vehicles. It uses headlights that consist of an array of LEDs.

Depending on the position of the oncoming car, some of these LEDs are automatically dimmed. In this way, while around the oncoming car is illuminated, the driver side is dimmed. The AFS uses image sensors to detect the position of the oncoming vehicle. Figure 1 shows how the AFS adjusts the headlights to dim the driver side of the oncoming

LITRETURE SURVEY

MeftahHrairi and Anwar, AbuBakar[1] presented the "Development of an Adaptive Headlamp system"**:** In this paper author described the steerable headlight by using components which are easily available in market which indirectly increase the reliability decrease product cost. They have used real time sensor to sense steering angle and vehicle speed. This paper introduces the AFS system that will modify the conventional headlamp which improves the reliability and safety. Potentiometer sensor is used to sense the steering angle of car. Pic microcontroller is used.Servo motor is used to move the headlight horizontally. This paper described only horizontal moving of the headlamp. Therefore system only achieve the 50% of actual adaptive headlamp working.If we actually develop this system then when the headlight at right most position or leftmost position we see the obstacle clearly than conventional headlamp this is the advantage of system another advantages are system has high adaptability, lesscost, simple and dependable assembly design. After studying the development of an adaptive headlamp system [1] author have described only horizontal moving of headlamp they have not described the vertical moving of headlamp.

YaliGuo,QinmuWu,Honglei Wang[2] presented the "Design And Implementation Of Intelligent Headlamp Control System Based On Can Bus" : In this paper author described motion model of headlights in which turn angle of headlamps in horizontal and vertical direction can be calculated by mathematical equations. System contains hardware as well as software. System achieves the dynamic adjustment for headlamp horizontally and vertically by receiving data from CAN bus.

The system uses the potentiometer to sense the vehicles steering angle and uses no of keys to simulate speed of car. Angle of the potentiometer sensor ranges from 0 degree to 180 degrees. It is observed that the time required for data processing is less than 10 millisecond so time delay between the potentiometer and headlight can be neglected.

After studying this paper[2] I have conclude that system is feasible and CAN bus is used as a data transmission channel. Advantages of system is fast response, stability but disadvantage is small error is present in the system because of mathematical calculations.

M.Canry,S.Cherfan,P.Lecocg,J.M.Kelada,A.Kemeny[3] presented the "Application of Real Time Lighting **Simulation for Intelligent Front Lighting Studies** "Inthis paper author introduce the adaptive headlight strategies which are used in Renault Automobile. They have use the real time lighting simulation. The driving simulator include driver cockpit, an image generator which provide realistic restitution of headlight distribution, a sound generator, realistic traffic and mam machine interface. This system can easily handled during switching in real time driving conditions. system is called as Valeo Lighting system Branch which is developed by Ranault Lighting Department which includes dynamically directing additional headlamp according to the behavior vehicle, environment and road conditions.

They have used the Halogen bulbs which provides 30% more light than other previousgeneration bulbs. Because of the clear lens and complex shape uses 80% light that are previously wested on shields and bezels. They have used Xenon lightning which increasing the performance in the range: double flux and beam width, 40% reduction in use of electricity, day light colour rendering,30% increased range. Ranault Xenonheadlamp has been installed in the safranein early 1997.

After studying this paper[3] We found that Author divide the light distribution in Motoway light, Adverse weather light, Town Light, Bending Light. Several simulation have been developed by Ranault Research Department since 1989.Lighting simulator save cost and time that are required for the headlight turning process.

ShirsathShashikant,Mechkul M.A.[4]presented the “Adaptive front Light System “In this paper author described the horizontal swings of headlamp by sensing steering angle and vertical swings of headlamp by sensing distance of incoming vehicle. Author mainly consider the Accuracy, Consistency, Availability of components.
The aim of this system is built cost effective Adaptive Front Light System that will achieve increase safety comfort and enhance drivers visibility. The system consist of input sensors, a FPGA as head of system and motor for rotating headlights. Author suggested to use the ultrasonic distance sensor and potentiometer as a steering angle sensor. Horizontal movement achieved through potentiometer sensor and Vertical movement is achieve through ultrasonic distance sensor. By taking reference from this paper I have used the ultrasonic distance sensor for vertical movement but for horizontal I have used the image processing which reduces time required for horizontal movement of headlamp this is the advantage of AFS system over this system

CIRCUIT DIAGRAM

The potentiometer is a strip of resistance material. It’s one terminal is connected to the anode and other terminal is connected to the cathode. The voltage output is drawn from a third terminal which has slide able contact with the resistive material. As the contact closes to anode end, the voltage output gets closer to supply voltage due to low resistance. When the contact slides away from the anode end, the voltage output is dropped due to a higher resistance between the anode and output terminal because the resistance is increased proportionally to the length of resistive material between them. At the steering end, the potentiometer is set to a mid value for default, as the steering rotates either direction, the contact of the potentiometer either slides away or closer to anode end and the output voltage from the potentiometer is reduced or increased with respect to the default value.

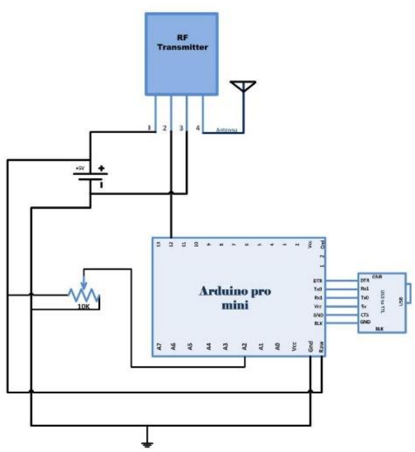

Fig a) : Transmitter of AFS

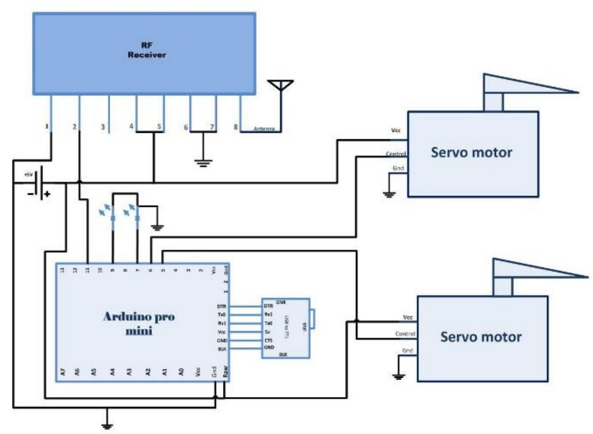

Fig b) : Receiver of AFS

The potentiometer is a strip of resistance material. It’s one terminal is connected to the anode and other terminal is connected to the cathode. The voltage output is drawn from a third terminal which has slide able contact with the resistive material. As the contact closes to anode end, the voltage output gets closer to supply voltage due to low resistance. When the contact slides away from the anode end, the voltage output is dropped due to a higher resistance between the anode and output terminal because the resistance is increased proportionally to the length of resistive material between them. At the steering end, the potentiometer is set to a mid value for default,

as the steering rotates either direction, the contact of the potentiometer either slides away or closer to anode end and the output voltage from the potentiometer is reduced or increased with respect to the default value.

The potentiometer's voltage output is read as analog data at the A2 pin of the transmitter side Arduino. The reading is converted to character form and transmitted over RF using the Virtual Wire library functions in the program code.

The potentiometer reading is detected at the RF receiver and serially out to the pin 11 of the receiver side Arduino. The program code at the receiver side Arduino detects the potentiometer reading by reading the transmitted character buffer and converting the reading to an integer value. The value is compared with a default calibrated value and the direction of turning is judged accordingly. By detecting the direction car has turned, PWM signals are passed to the RC servos for rotating along the direction of turning. The LEDs remain ON by default and rotates along with the servo as they are mounted on them. Check out the transmitter side Arduino code to learn how analog reading from the potentiometer is detected in the code and converted to character form and serially transmitted over RF. Then, check out the receiver side Arduino code to learn how character buffer is read and interpreted to an integer value and based on the value read, the servos are operated.

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BLOCK DIAGRAM

Adaptive headlights are an active safety feature designed to make driving at night or in low-light conditions safer by increasing visibility around curves and over hills. Proposed system controls two operations: One is to control the angle of the headlight and another is to control the intensity of the headlight.

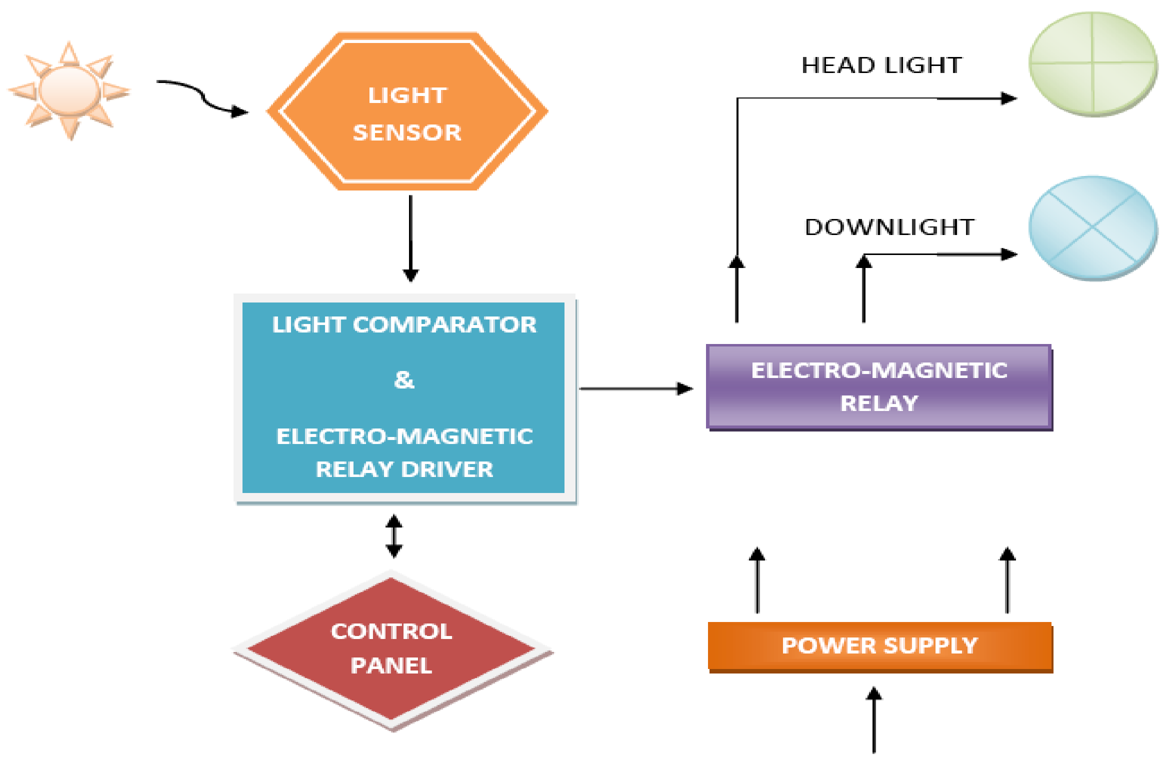

Fig : Block Diagram of AFS

When driving on the curve road, standard headlights continue to shine straight ahead, illuminating the side of the road and leaving the road ahead of you in the dark. Adaptive headlights, on the other hand, turn their beams according to your steering input so that the vehicle's actual path is lit up. This is achieved by using single turn potentiometer as steering input which senses the steering wheel position. The voltage value of the pot is read by Arduino Uno R3, a 8-bit microcontroller and thus according to the Arduino output corner and deep corner LEDs are turned ON. Similarly during a night drive in a highway, driving right side of a one way has become too hard mainly because of the glare that is coming from a vehicle coming in the opposite direction, this may sometimes cause the driver to crash to the vehicle in front of him, switching to low beam is the solution for it but people tend to forget it. So in our project we intend to automate this by measuring the intensity values from the opposite vehicle using LDR and sensing the distance between incoming vehicle and subject vehicle using ultrasonic sensor in turn reducing accidents.

SYSTEM SPECIFICATION

An adaptive vehicle lighting system refers to a set of advanced lighting technologies in automobiles designed to optimize visibility and safety while driving in various conditions. Here's a breakdown of the typical specifications and features you might find in such a system:

1)Automatic Adjustment:

The system automatically adjusts the intensity and direction of the vehicle's headlights based on factors such as vehicle speed, weather conditions, road type, and surrounding traffic.

2)Dynamic Range Control:

This feature adjusts the brightness of the headlights based on the distance of the vehicle ahead or oncoming traffic to prevent glare and maximize visibility without blinding other drivers.

3)Adaptive Headlights:

These headlights can swivel or pivot in response to steering input, allowing the light to follow the path of the road as the driver turns the wheel. This improves visibility around curves and corners.

4)Sensor Integration:

Adaptive lighting systems often incorporate sensors such as cameras, lidar, radar, and ambient light sensors to gather real-time data about the vehicle's surroundings and driving conditions, enabling precise adjustments to the lighting.

5)Adaptive Front Lighting System (AFS):

AFS adjusts the direction and range of the headlights based on factors such as vehicle speed, steering angle, and weather conditions to provide optimal lighting for different driving scenarios.

SOFTWARE DEVLOPMENT

On the transmitter side Arduino, first, the program code imports the required standard libraries. The Virtual Wire library is required to read the analog reading from pin A2. Global variables led Pin and Sensor1Pin are declared and mapped to Pin 13 where transmission progress indicator LED is connected and Pin A2 where output pin of the potentiometer is connected respectively. A variable Sensor1Data is declared to store potentiometer voltage reading. A character array Sensor1CharMsg is declared to stored decimal representation of the potentiometer reading.

```
#include <VirtualWire.h>
// LED’s
const int ledPin = 13;
// Sensors
const int Sensor1Pin = A2;
// const int Sensor2Pin = 3;
int Sensor1Data;
//int Sensor2Data;
char Sensor1CharMsg[4];
```

A setup() function is called where indicator LED pin is set to output while sensor connected pin is set to input mode using pinMode() function. The baud rate of the Arduino is set to 9600 bits per second using the serial.begin() function. The baud rate for serial output is set to 2000 bits per second using the vw_setup() function of the VirtualWire library.

A loop() function is called where the analog reading (in terms of voltage measurement at the pin) is fetched using the analogRead() function and stored in “Sensor1Data” variable. The potentiometer reading (an integer value) is converted to the character of base 10 using itoa() function and stored in Sensor1CharMsg array.

The potentiometer reading is serially out as human-readable ASCII text using the Serial.print() and Serial.println() function.

The transmission progress indicator LED is switched on by passing a HIGH to pin 13. The character message containing the potentiometer reading is sent serially using the vw_send() function and vw_wait_tx() is used to block transmission until a new message is available for transmission. The LED at pin 13 is switched OFF by passing a LOW to indicate successful transmission of the message.

This ends the transmitter side Arduino code.

On the receiver side Arduino, the program code first imports the required standard libraries. The VirtualWire library is imported to receive potentiometer reading from RF. The pins where servos are connected are mapped to “servo” and “servo1” variables and other global variables – “angle” to represent the angle of rotation for servos and “PWM” for respective pulse width duration are declared. The “Sensor1Data” is declared to store integer value of reading and Sensor1CharMsg array is declared to hold character representation of reading through RF.

A setup() function is called where baud rate of the Arduino is set to 9600 bits per second. The servo and LED connected pins are set to digital output using the pinMode() functions. The pins where LEDs are connected are set to HIGH to keep the LEDs switched ON by default.

The RF transmitter and receiver module does not have Push To Talk pin. They go inactive when no data is present to transmit or receive respectively. Therefore vw_set_ptt_inverted(true) is used to configure push to talk polarity and prompt the receiver to continue receiving data after fetching the first character. The baud rate for serial input is set to 2000 bits per second using vw_setup() function. The reception of the data is initiated using vw_rx_start().

A loop() function is called inside which, array “buf[]” to read serial buffer and “buflen” variable to store buffer length are declared.
The character buffer is detected using the vw_get_message() function if it is present, a counter “i” is initialized. The character buffer is stored in the Sensor1CharMsg array using the for loop with the initialized counter.
The null character is detected in the buffer stream so that no garbage value gets stored. The characters received from the buffer are converted to the integer value and stored in “Sensor1Data” variable.

The variable value along with relevant strings enclosed is passed to the microcontroller’s buffer and Sensor1Data value is passed to servoPulse and servoPulse1 custom functions as a parameter to rotate servos.

In servoPulse function, the Sensor1Data which has been passed as angle parameter is taken and multiplied by 11 then added to 1000 to get pulse width duration in microseconds. The servo connected pin is set to HIGH and delay of calculated duration is passed. The HIGH output to servo connected pin is then terminated.

The same logic as implemented in servoPulse function is implemented as it is in servoPulse1 function.

This ends the receiver side Arduino code.

PCB DESIGN

There are two circuits in the project – Steering Control Input Circuit and Headlight Servo circuit. In the steering control input circuit, a 10K ohm potentiometer is connected to A2 analog pin of the Arduino Pro Mini. The other two terminals of the potentiometer are connected to VCC and ground. The Arduino is interfaced to RF transmitter by connecting the serial input pin (Pin 2) of the RF transmitter to pin 12 of the Arduino. An antenna can be used at the pin 4 of RF transmitter to improve transmission signal, however, it is optional and not really needed as the operational range of the project will be limited to the front dimensions of the car. On life-size implementation of the project, some mechanical assembly needs to be designed so that the potentiometer knob rotates with respect to the rotation of steering wheel.

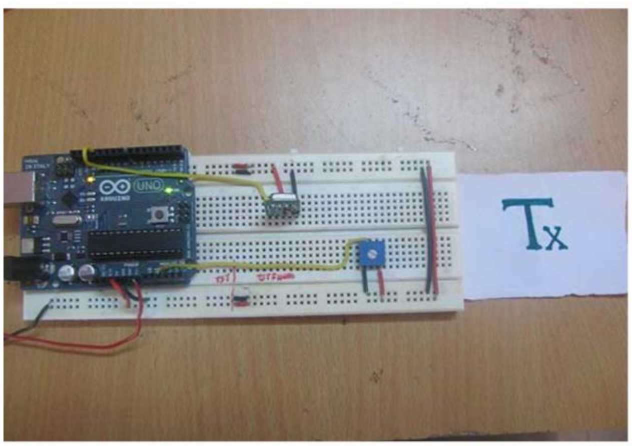

Fig a) Transmitter circuit

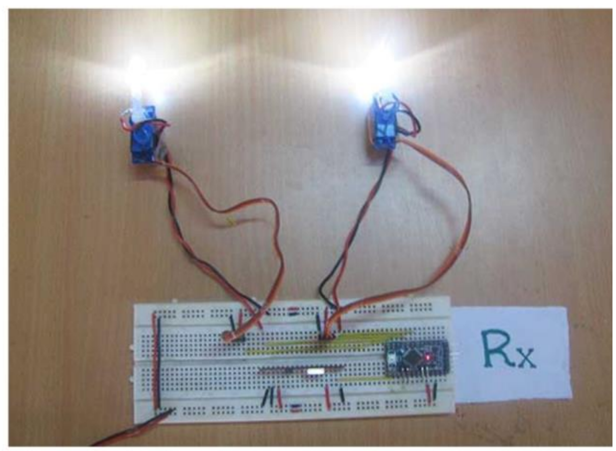

Fig b) Reciver circuit

At Servo circuit, an RF receiver is connected to the another Arduino board by connecting serial out pin (Pin 2) of the receiver module to pin 11 of the Arduino. Again an antenna can be used with the RF receiver which is optional only to improve signal strength. There are two LEDs connected at pin 9 and 7 of the Arduino which are assumed the car headlights in the circuit. These LEDs are mounted on the Servo motors. There are two RC servo motors connected to data pins 5 and 6 of the Arduino with respect to the headlight representing LEDs.

APPLICATION

When speaking of actuators or moving mechanisms in the automotive industry, headlights and lighting systems are an unavoidable topic. Static headlights with light bulbs are now a thing of the past. Nowadays, manufacturers do their best efforts to improve driving comfort, and in this regard, **automatic adaptive headlights** are increasingly gaining ground. Do you know how those motions are achieved? And which benefits they offer when driving?

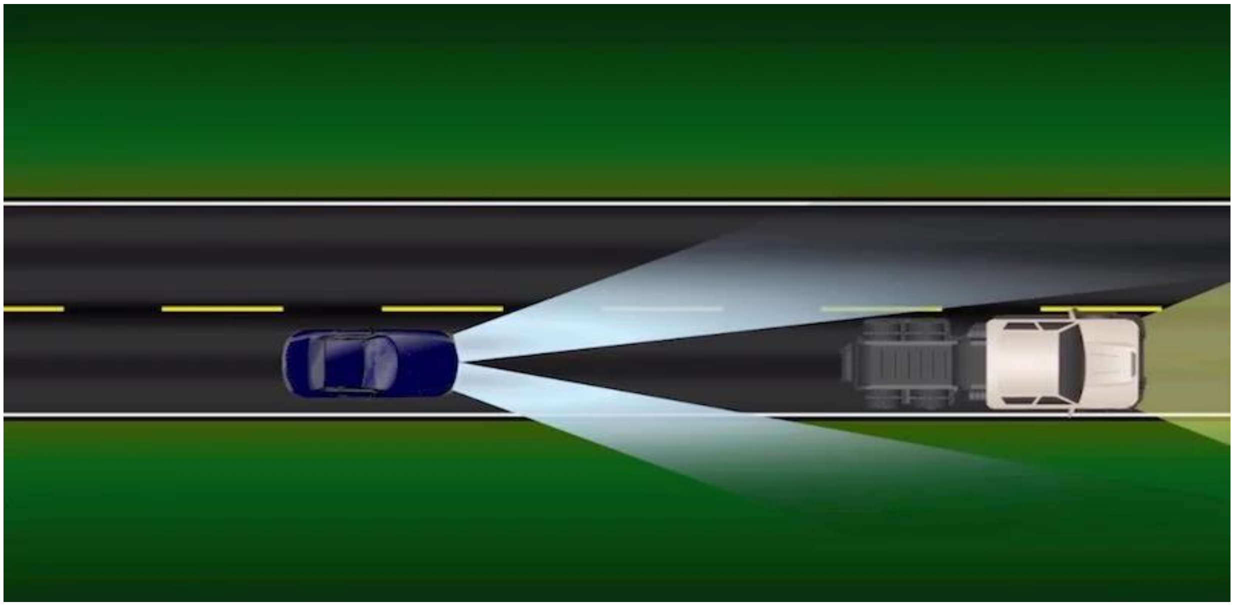

Fig : Shows The implementation of AFS

1. They allow for safer driving

Automatic adaptive headlights is one of the innovations of **automotive engineering** that is destined to improve **driving safety**. These systems allow automatic lighting on the road without the need for driver interaction. The usefulness of these adaptive headlights involves the all-important task of preventing accidents.

Imagine you are driving at night along a two-lane road with no lighting, and you approach a curve. Even if you are driving at the proper speed, visibility is reduced. Thanks to these adaptive headlights, lights automatically rotate depending on the angle of rotation of the steering wheel on every curve, offering a better visibility over the road ahead. In 2006, for example, over 46% of fatal accidents took place at night. **Adaptive headlamps** enhance the night driving experience, making the trip safer.

2. They avoid blinding other vehicles or pedestrians

Adaptive headlights not only benefit the driver, but also other vehicles. For example, if travelling along a two-way sloped road, the light pattern is reduced to prevent glare on oncoming drivers.

3. They adapt better to driving conditions

Automatic adaptive headlights have high and low beams which are automatically adjusted by a set of sensors. Next, information is sent to a computer regarding driving conditions and traffic. They have systems that are capable of detecting oncoming traffic and the taillights of cars in front of us. Based on this information, headlights are automatically turned on. The driver will be able to use the highest beam without blinding other drivers.

4. They improve driving in adverse weather

Driving in the rain or fog increases road accident rates. One of the factors is lack of visibility. Adaptive headlights can help the driver even in adverse weather. For example, this type of system activates anti-fog lights, and at a rate of speed slower than 70 km/h, they rotate the left headlight about 8 degrees outwards and 1 degree down. Thus, glare in foggy weather is diminished and the left side is more illuminated.

ADVANTAGES AND DISADVANTAGES

Advantages:

They look amazing, but their design serves one purpose only, your safety. Adaptive headlights move up and down, left and right, to enhance the driver's field of view in low-light conditions. They are a great feature we can find on many motor vehicles nowadays. Why are they there, and how exactly do they work? Let's shed some light on this subject.

1. **Improved visibility**:
   Adaptive lighting systems adjust the direction and intensity of the headlights based on driving conditions, such as speed, steering angle, and weather. This enhances visibility for the driver, especially during night-time driving or in adverse weather conditions like fog or rain.

2. **Enhanced safety**:
   By providing better illumination of the road ahead and potential obstacles, adaptive lighting systems contribute to safer driving experiences. They help drivers to detect pedestrians, animals, or other vehicles sooner, reducing the risk of accidents.

3. **Reduced glare**:
   These systems can automatically dim or adjust the headlights when approaching other vehicles to prevent blinding oncoming drivers, enhancing overall road safety and comfort for all road users.

4. **Extended bulb lifespan**:
   Adaptive lighting systems often use technologies like LED or Xenon bulbs, which have longer lifespans compared to traditional halogen bulbs. This can result in reduced maintenance costs and fewer instances of bulb replacement.

5. **Customizable lighting modes**:
   Some adaptive lighting systems offer customizable lighting modes or settings, allowing drivers to tailor the lighting to their preferences or driving conditions, such as highway driving, city driving, or off-road conditions.

6. **Increased energy efficiency**:

Adaptive lighting systems can optimize energy usage by only illuminating specific areas of the road as needed, rather than keeping all headlights at full brightness continuously. This can result in energy savings and potentially improve fuel efficiency in vehicles.

7. **Enhanced aesthetics**:
   Beyond their functional benefits, adaptive lighting systems can also enhance the aesthetics of vehicles. Sleek, modern lighting designs contribute to the overall attractiveness and appeal of a vehicle, potentially increasing its marketability.

8. **Technological innovation**:
   Adaptive lighting systems represent technological innovation in the automotive industry, showcasing a vehicle manufacturer's commitment to advancing safety, comfort, and efficiency features in their products. This can attract tech-savvy consumers and differentiate vehicles in a competitive market.

Disadvantages :

While adaptive vehicle lighting systems offer numerous advantages, they also come with some potential disadvantages:

1. **Complexity and cost**:
   The technology behind adaptive lighting systems can be complex, involving sophisticated sensors, actuators, and control algorithms. This complexity can increase the manufacturing cost of vehicles equipped with these systems, potentially making them more expensive for consumers.

2. **Maintenance and repair costs**:
   The intricate nature of adaptive lighting systems may lead to higher maintenance and repair costs compared to traditional lighting systems. Components such as sensors, motors, or control modules may require specialized diagnostics and servicing, which could be more costly and time-consuming.

3. **Reliability concerns**:
   Like any electronic system, adaptive lighting systems are prone to malfunctions or failures. Sensor errors, software glitches, or mechanical issues could compromise the functionality of the system, leading to reduced performance or safety concerns for drivers.

4. **Compatibility with aftermarket modifications**:
   Aftermarket modifications or accessories, such as aftermarket headlights or suspension modifications, could potentially interfere with the operation of adaptive lighting systems. This may limit the options available to vehicle owners who wish to customize their vehicles.

5. **Limited effectiveness in certain conditions**:
   While adaptive lighting systems excel in providing improved visibility and safety in most driving conditions, they may have limitations in extreme weather conditions such as heavy snowfall or dense fog. In such situations, visibility may still be compromised despite the adaptive features.

6. **Power consumption**:
   Although adaptive lighting systems are designed to optimize energy usage, they still require power to operate. Continuous use of adaptive features, particularly in urban driving conditions with frequent stops and starts, could lead to increased power consumption and potentially affect fuel efficiency.

7. **Complexity for drivers**:
   Some drivers may find the operation of adaptive lighting systems to be overly complex or confusing. Understanding how the system works and how to adjust its settings effectively may require additional time and effort, potentially leading to user frustration or dissatisfaction.

RESULT ANALYSIS

An analysis of an adaptive vehicle lighting system would typically involve several key aspects:

1. **Performance Evaluation**:
   This includes assessing how well the lighting system adapts to different driving conditions such as varying weather conditions, road types, and ambient lighting. It involves testing the system's ability to automatically adjust the intensity, direction, and distribution of light to optimize visibility without causing glare for other road users.

2. **Safety Impact**:
   Analyzing the impact of the adaptive lighting system on safety is crucial. This involves studying accident data before and after the implementation of the system to determine if there's a reduction in accidents or near-misses, particularly in low visibility conditions.

3. **User Experience**:
   Gathering feedback from drivers about their experience with the adaptive lighting system is essential. This includes aspects such as ease of use, comfort, and overall satisfaction with the system's performance.

4. **Energy Efficiency**:
   Assessing the energy efficiency of the adaptive lighting system is important to understand its impact on fuel consumption or battery life in the case of electric vehicles. This involves measuring the power consumption of the system under different operating conditions.

5. **Cost-Benefit Analysis**:
   Evaluating the cost-effectiveness of implementing the adaptive lighting system compared to traditional static lighting systems. This includes considering the initial cost of installation, maintenance costs, and potential savings from improved safety and energy efficiency.

6. **Regulatory Compliance**:

   Ensuring that the adaptive lighting system meets all relevant regulatory standards and requirements for vehicle lighting is crucial. This involves testing the system to ensure compliance with lighting regulations in different regions.

7. **Market Analysis**:

   Understanding the market demand for vehicles equipped with adaptive lighting systems and analyzing the competitive landscape can provide insights into the potential adoption and penetration of the technology.

CONCLUSION

At night, driving a vehicle on cornering of road can be quite challenging. The straight illumination of the headlamps makes it difficult to accommodate for sharp turns. Lack of visibility at turns in narrow roads can prove to be fatal collision of vehicle.

The system is inexpensive, simple and dependable assembly. This proposed model can be used in small and medium level cars also, which reduces the accident rate by illuminating the blind spots and eliminating the glaring effects due to oncoming vehicle by headlight angle and beam control mechanisms. This system is cost efficient, and it can be proven to be even more effective as it automatically dims its light when a vehicle comes at a closer distance, and thus, providing a better vision to the person.

Adaptive headlight system thus can be used as accessory in all running vehicles for proper illumination of road according the driving situation. This ensures higher degree of active safety in vehicles and assistance to driver.

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